The Stars on the Plate

Pedro Chiara Zina

Analyzing Global Michelin Restaurants

Goal

Explore how cuisine type, geography, and pricing relate to Michelin star ratings.

Dataset Used: three datasets merged into one master table

- ⭐ 1-Star Restaurants
- ⭐⭐ 2-Star Restaurants
- ⭐⭐⭐ 3-Star Restaurants

Enriched with:

- Average price per restaurant (source : Michelin Guide; The restaurants' price ranges index);
- Online review information (API).

Project Overview:

Hypotheses :

1. **Cuisine hypothesis:** Some cuisines are more likely to receive 3 stars (e.g., French).
2. **Geography hypothesis:** Large cities have more 3-star restaurants.
3. **Price hypothesis:** Average price increases with star level (1→3).
4. **Michelin stars:** Star level does not always reflect public opinion.

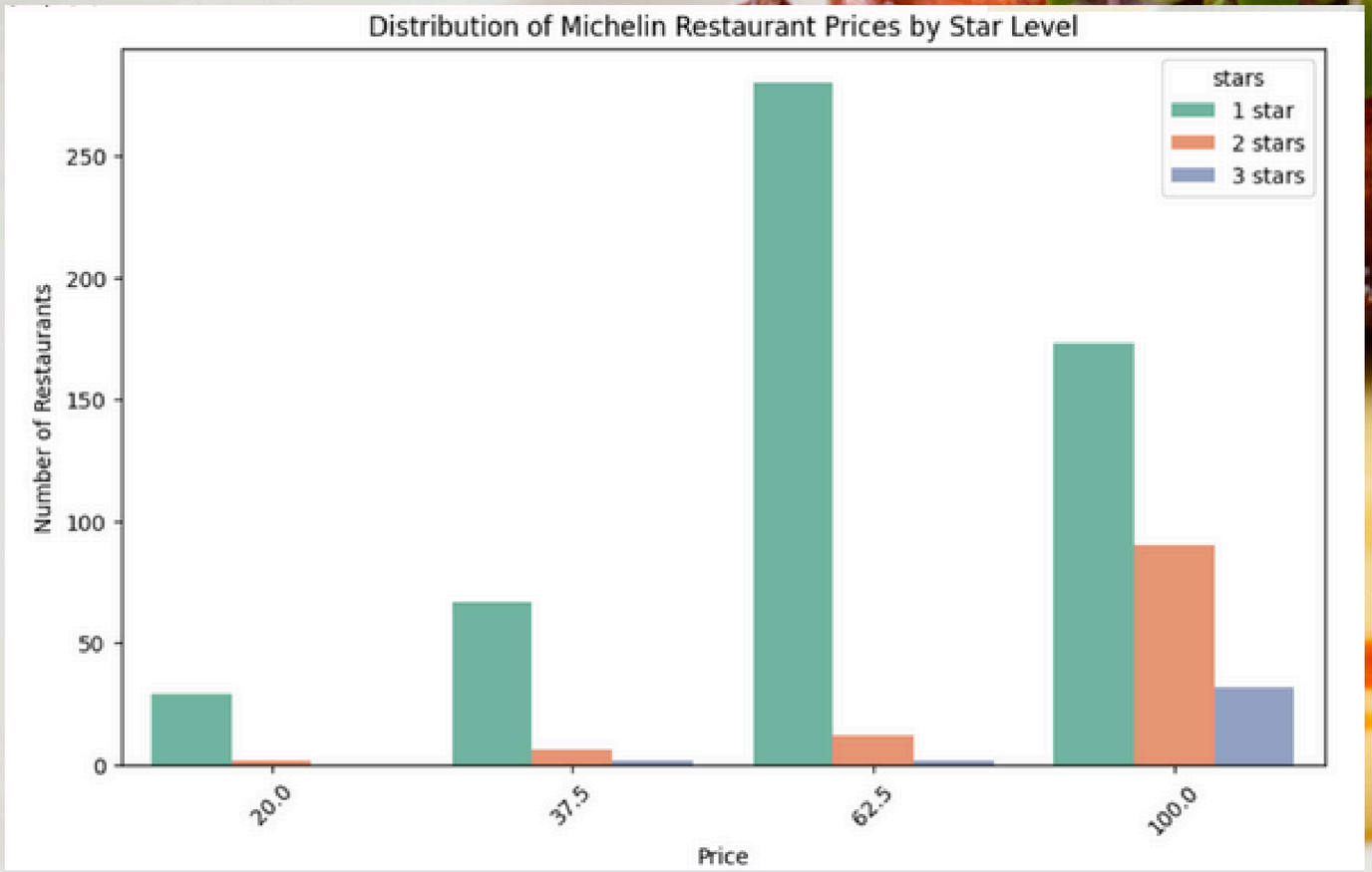
Data Wrangling and Cleaning

- **Data Integration:** Combined 3 databases through concatenation;
- **Special Characters:** Used 'unidecode' to clean restaurant names of accents and special characters;
- **Cuisine Simplification:** Reduced over 100 cuisine types into ~10 coherent groups;
- **Geographic Organization:** Grouped and ordered suburbs by city and region for clarity;
- **Symbol Inconsistencies:** Fixed entries with too many $ symbols, removed extra spaces and hidden characters;
- **Converting Dollar Signs to Average Price**: $ → 20, $$ → 37.5, $$$ → 62.5, $$$$ → 100 ;
- **Handeling Missing Prices:** Filled missing values using median price per Michelin star level.

Exploratory Data Analysis

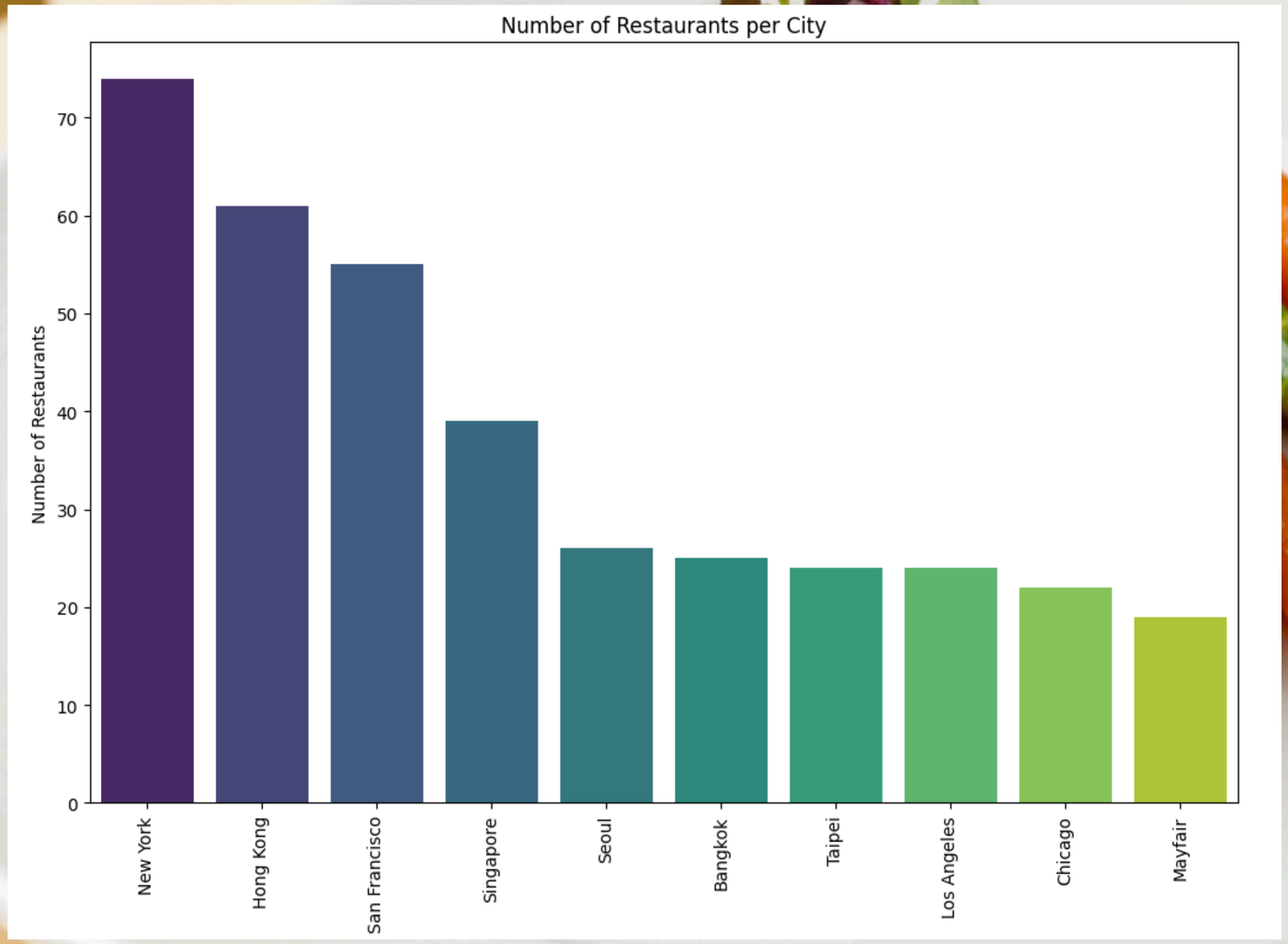
As we can see:

1-star restaurants dominate the lower price ranges, while **2-star restaurants** appear more frequently in mid-to-high price ranges. **3-star restaurants** are mostly concentrated in the highest price category, confirming that the most prestigious restaurants tend to be the most expensive.



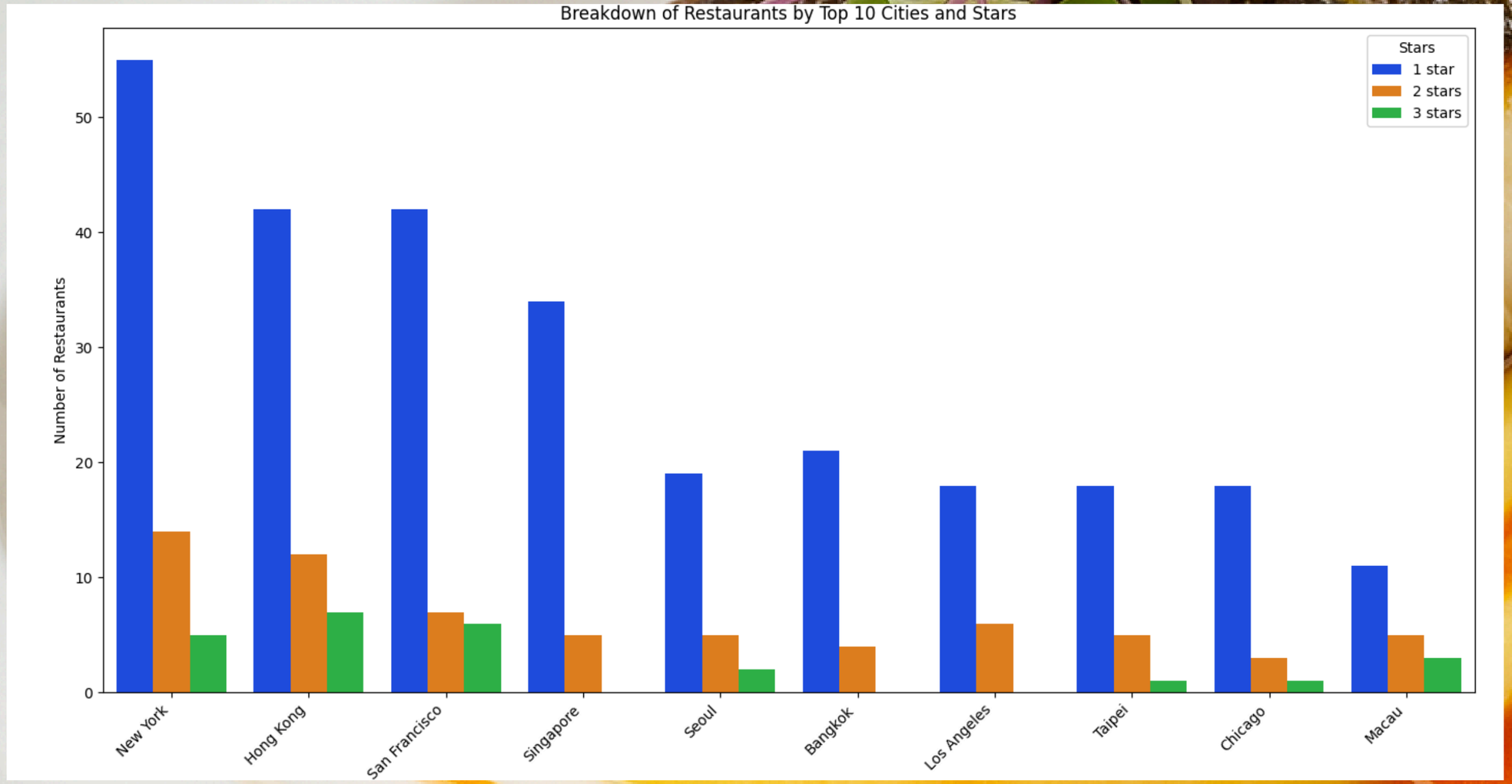
Where are Michelin-starred restaurants distributed globally?

New York, Hong Kong, and San Francisco are major culinary and cultural hubs that attract Michelin-star.



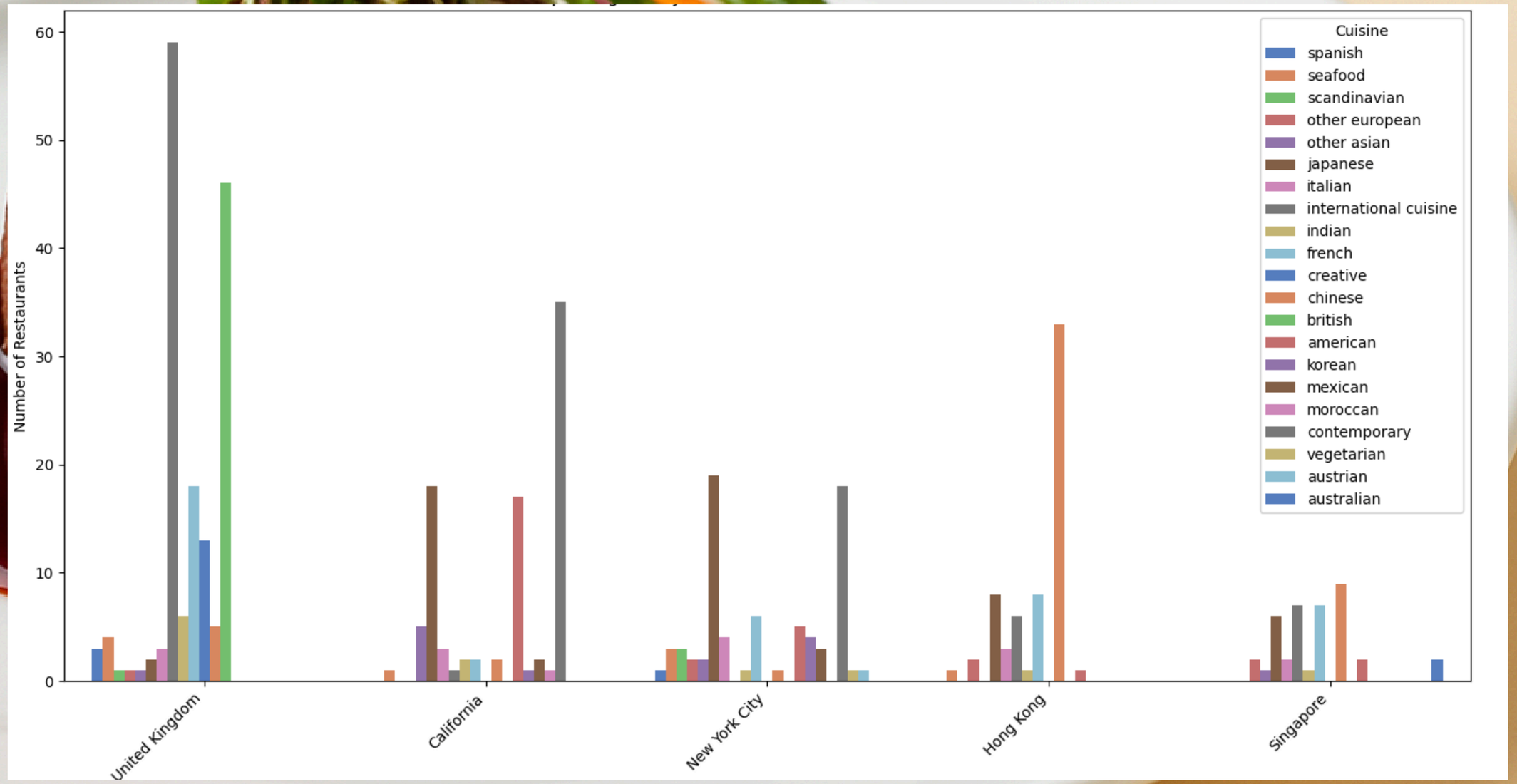
What is the distribution of 1, 2, and 3-star Michelin restaurants in the top 10 cities globally?

New York leads in 1-star and 2-star Michelin restaurants, offering a diverse array of high-quality dining. Hong Kong, however, excels in 3-star Michelin establishments, indicating a focus on elite dining experiences.

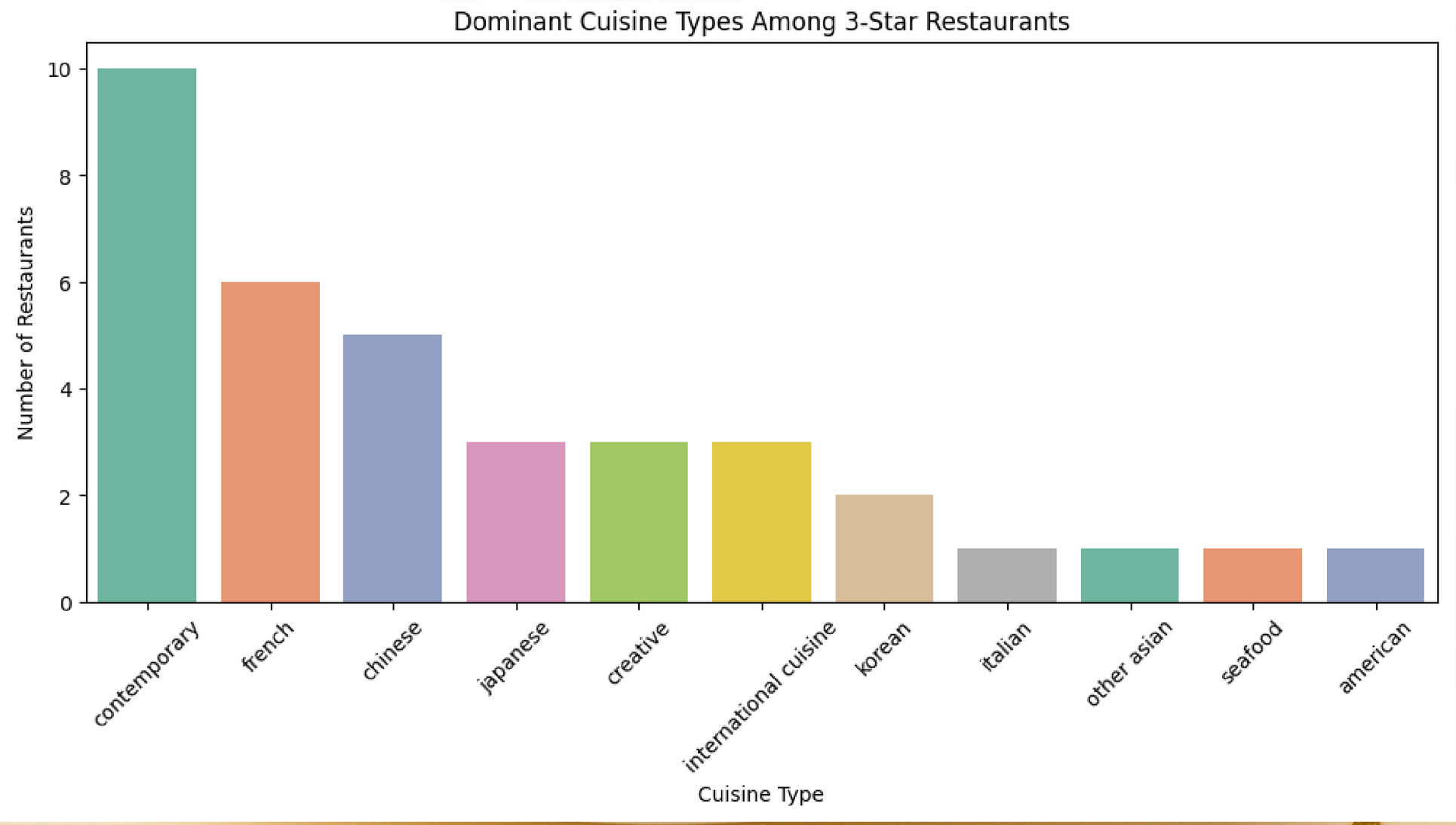


Which cuisine styles are most prevalent among Michelin-starred restaurants in the top 5 regions?

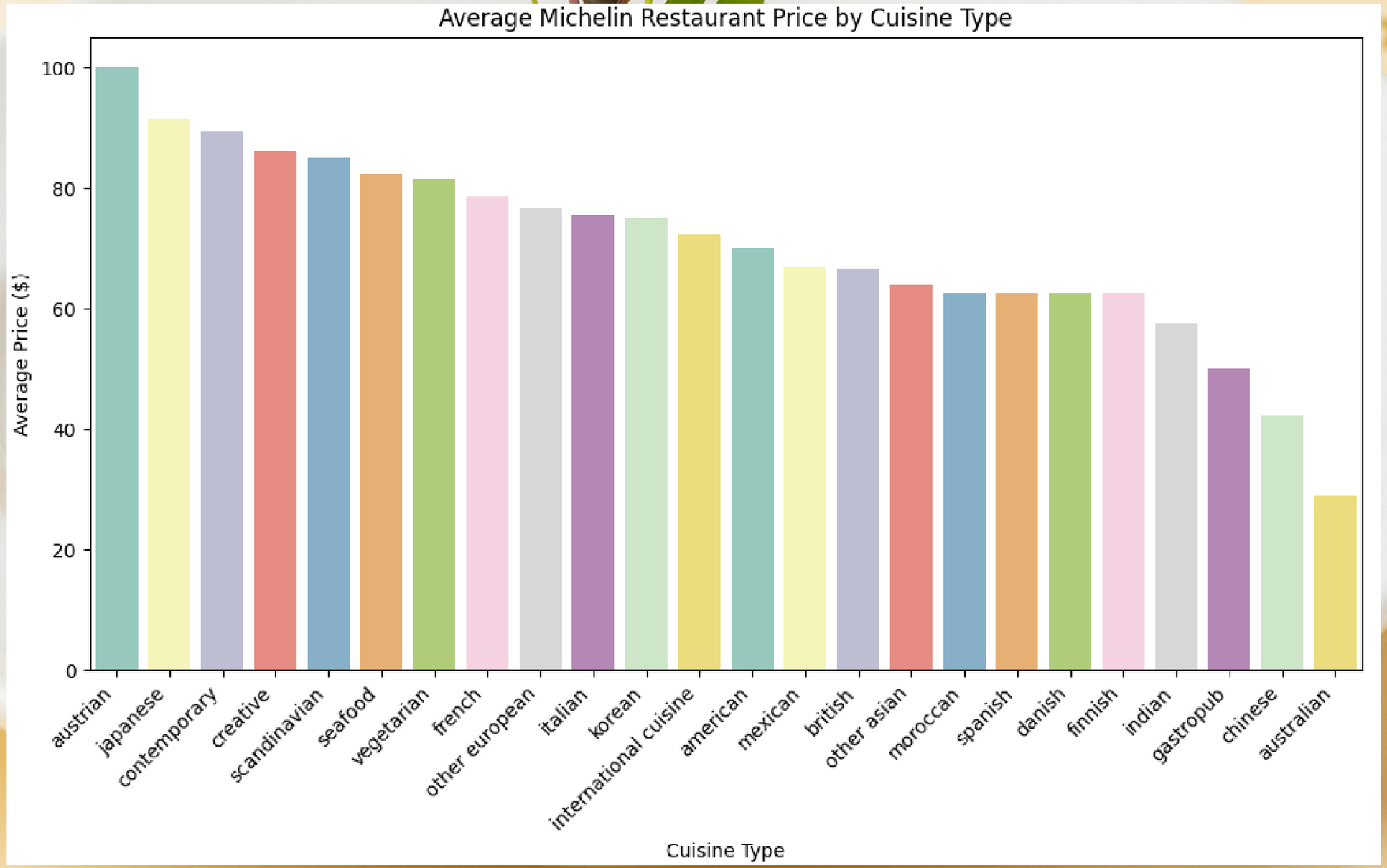
Contemporary emerges as the most common choice, followed by regional specialties unique to each area.



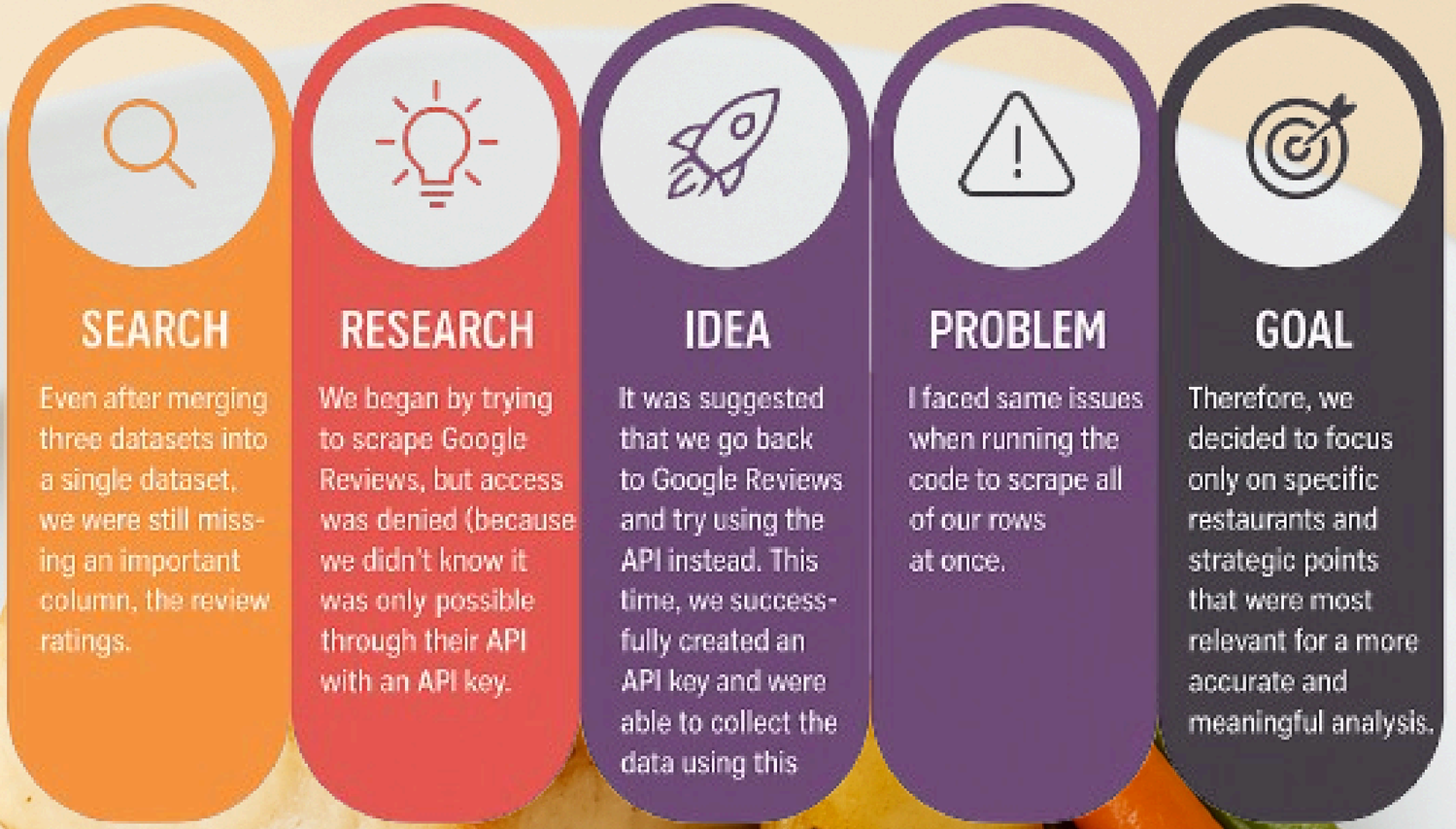
Among 3-star Michelin restaurants, the dominant cuisines are contemporary, French, and Chinese.



Exploratory Data Analysis



Webscraping & API



	name	review_rating	review_count
109	chef's table at brooklyn fare	4.2	574.0
162	eleven madison park	4.4	3149.0
448	per se	4.5	2009.0

Takeaways

Pricing Strategy:

- 1-Star: Dominates lower price ranges.
- 2-Star: Thrives in mid-to-high price points.
- 3-Star: Found in the highest price category.

Top Locations:

- New York: Ideal for 1-star and 2-star goals.
- Hong Kong: Focus for 3-star prestige.
- San Francisco: Another major hub for Michelin stars.

Culinary Trends:

- Common Choices: Contemporary cuisine is broadly favored and scored 4.5 /5 customer ratings on Google reviews
- 3-Star Dominance: Contemporary, French, and Chinese.

Cuisine Costs:

- Most Expensive: Austrian
- Followed by: Japanese, Contemporary & Creative, Scandinavian, French, Italian.

Our strategy:

We aim to enter the market with 1 star price ranges and then scale up after a first market test.

Seoul presents a promising opportunity for further investigation, given low competition in 1 star Michelin. With its strong GDP and a vibrant, youthful population the city is an attractive market for aspiring restaurateurs.

Contemporary cuisine is broadly favored among Michelin-rated restaurants. Therefore, we want to specialize in French contemporary cuisines incorporating modern cooking methods.

Thank You!

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Chiara
Zina